

Development of Low-Cost Real-Time Data Acquisition System for Process Automation and Control

Mohammad Haizad, Rosdiazli Ibrahim, Anas Adnan, Tran Duc Chung, Sabo Miya Hassan

Department of Electrical and Electronic Engineering,

Universiti Teknologi PETRONAS,

32610, Bandar Seri Iskandar, Perak, Malaysia

haizadtarik@gmail.com, rosdiyalzi@petronas.com.my, anas.adnan.21@gmail.com, chungtran@acm.org, hsmiya2010@gmail.com

Abstract—The use of data acquisition system (DAQ) is important in process industry as it enables analog signal from physical system to be converted into a digital signal for processing by computer and vice versa. However, many of the DAQs available in the market today are complex and expensive. This paper presents the development of a simple, low-cost and open source microcontroller-based DAQ system for process automation and control. The proposed DAQ system is built around Arduino Mega 2560 microcontroller. The test experimental results of the developed DAQ system proved its suitability for application in process monitoring and control applications.

Keywords—low-cost; data acquisition system (DAQ); industrial; process; automation; control

I. INTRODUCTION

The need for sensors in the process automation industries is increasing at a very fast rate [1]. This is due to the development of plants requiring a higher degree of automation [2], in which sensors are needed to feed the control system of the plants with crucial information. The traditional solution to operate a process plant that depends on analog technologies are going obsolete as end users prefer a more innovative measurement techniques and digital technologies [2]. The integration of digital technologies in process plant requires a more sophisticated plant to be designed. These plants will have numerous numbers of digital instruments installed to gather information for automation leading to an abundance of digital data flooding the system. A way to manage these data is through data acquisition (DAQ). DAQ is the core of any digital system to interface with the real world. These DAQs allow for the process signals from measurement devices to be gathered, processed and analyzed for the purpose of control, storage, presentation, etc. [3,4].

There are generally three categories of commercial DAQ systems in the industries [5]. The first category of the DAQ is the embedded microcontroller-based system [5]. This system has the advantage of portability as it is based on the autonomous instruments. The fixed architecture of the embedded design also increases the robustness of the DAQ and its tolerance to fault such as short circuits [6]. This type of

DAQ is suitable to be used in a hazardous environment. The second category is the modular acquisition system that is connected to a specialized local bus like PCI eXtensions for Instrumentation (PXI). This DAQ can be more powerful than the embedded system. It is also suitable to be used in an environment that has a need for future growth [6]. The last category is a computer-based, this type of DAQ is connected to a monitoring computer where it uses the computer processing power [6]. The DAQ will act as a signal conditioner before the data is processed by the connected computer to perform the desired data manipulation, storage or visualization. The computer-based DAQ exists in two forms, external and internal modules. The internal module is an extension card that is connected to a computer's Peripheral Component Interconnect (PCI) slot. Meanwhile, the external module is a separate module that is connected to a computer via the universal serial bus (USB). This type of computer-based DAQ is tethered to a computer which means that its processing is dependent on the computer.

DAQs have been used in a wide range of applications such as in biomedical industry [6,7] and it can also be used as a teaching tool [8]. In an academic environment, DAQ can be one of the teaching tools that can help to demonstrate practicality behind the theories. The demand for a low-cost DAQ is high especially for the academic purpose [9] however the current computer based DAQ available in the market are expensive and cumbersome [3,5,6,10]. A typical example is the multifunction DAQ USB-6001 by National Instrument, a low-cost multifunction DAQ for basic measurement [11]. Another example is the CB-7021 single-channel analog voltage or current output module manufactured by Measurement Computing [12]. A further comparison of the cost and basic functionalities of these commercial DAQs is shown in Table I.

In addition to being expensive, many of the DAQs that are currently available in the market are not open source and thus require proprietary software to operate. The types of software used in DAQs are usually expensive and have proprietary issues. An open source design project is a project that is published under the open source license. This means the procedure, circuit schematics diagram and design of both hardware and software can be copied and modified for redistribution freely without restriction [13]. Open source

hardware and software enables the fast prototyping as they are affordable, have a large community of users, hence getting support is easier [14]. An open source DAQ system allows the sharing of the well-proven system and provides the opportunity for future improvement [15].

Apart from the commercially available DAQ, a few research have been done to develop low-cost DAQ for monitoring and control. An Arduino based DAQ [16] and DAQ based on Atmega32 with open source companion software [17] has been developed used for temperature monitoring and control. In addition, an advance DAQ and Processing (DAQP) was also designed and implemented on field-programming-gate-array (FPGA), this helps the DAQ to have the same performance as commercial DAQ at a fraction of its cost [18].

This paper presents a low-cost real-time data acquisition system based on open source microcontroller which can be used for process automation and control. The basic specification of the proposed DAQ as compared to the commercially available DAQ are shown in Table 1. The rest of this paper is organized as follows: Section II details the methodology of developing the proposed DAQ and Section III presents the experimental result to verify the functionality of the proposed DAQ.

TABLE I. COMPARISON OF SEVERAL LOW-COST DAQ SYSTEMS

DAQ	Specification	Price
NI USB-6008	8 analog inputs (14-bit, 20 kS/s) 2 analog outputs (14-bit, 10 kS/s) 12 digital I/O Bus-powered for high mobility Compatible with proprietary software such as LabVIEW and NI Measurement Studio	High
Measurement Computing CB-7021	1 analog inputs (12-bit) 1 analog outputs (12-bit) Voltage or current output Compatible with proprietary software such as DASyLab@Software	High
Proposed DAQ System	16 analog inputs (10-bits, 10 kS/s) One analog output (12-bits, 50 kS/s) 53 digital I/O Open source hardware Compatible with Arduino IDE for monitoring and control.	Low

II. METHODOLOGY

This section presents the detailed procedures to develop a low-cost DAQ system based on an open-source Arduino Mega 2560 microcontroller. The hardware setup, tools, and software needed for the design as well as the testing procedure are explained in detail in this section.

A. The Proposed Low-cost DAQ System

The general hardware setup of the proposed DAQ system is shown in Fig. 1. The proposed DAQ system is built based on Arduino Mega 2560 microcontroller. The system is connected to a host PC via serial communication. It enables the proposed system to send and receive information to/from the computer

or specifically the MATLAB software and Arduino IDE. The DAQ will be powered using a USB cable connected to the PC.

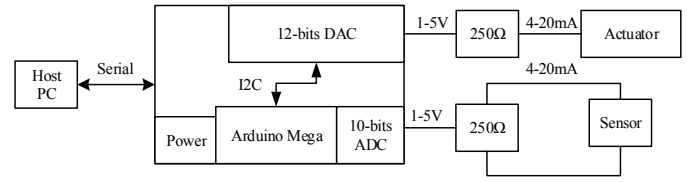


Fig. 1. Hardware setup of the proposed DAQ

The Arduino Mega 2560 is interfaced with a digital-to-analog converter (DAC) to enable it to give analog output. This enables the system to have a single 12-bits analog output. The Arduino Mega 2560 communicate with the DAC via Inter-Integrated Circuit (I2C) interface. The built-in Arduino ADC will enable the system to have 16 analog inputs. This allows the proposed system to monitor 16 different instruments. The detailed pin connection for the proposed system is tabulated in Table II.

TABLE II. ARDUINO INTERFACE PINOUT TABLE

Arduino	Connection
5V	DAC-VCC
GND	DAC-GND
D20	DAC-SCL
D21	DAC-SDA
A0 to A15	Analog Sensor Input

B. Component Selection

The selected software for this work is the Arduino IDE. Arduino IDE is used to load the developed firmware into the Arduino board and for control and monitoring of the DAQ board. It also can perform real-time monitoring of the board using the serial port monitor. The selected microcontroller board is Arduino Mega 2560 which is a widely used microcontroller. This is because it is affordable and has many analog inputs pins number which will enable the proposed DAQ system to be able to receive inputs from many transmitters. In order to perform control signal using the Arduino board, an additional DAC, i.e., MCP4725, is used since the microcontroller does not have a digital-to-analog converter (DAC). Both the MCU and the DAC's basic specifications are presented in Table 2.

TABLE III. MCU AND DAC BASIC SPECIFICATIONS

Device	Basic Specification	Detail
Arduino Mega 2560	ADC	10 bits
	Processing Speed	16 MHz
	Supported Interface	UART, I2C & SPI
DAC MCP4725	Resolution	12-bits
	Supported Interface	I2C
	Operating Voltage	3.3 – 5V

C. Experiment Procedure

To verify the functionality of the proposed low-cost DAQ system a test experiment is conducted. The output of the DAQ is used as the input for the DAQ to represent reading from the instrument. The procedure to conduct the testing experiment is shown in Fig. 2.

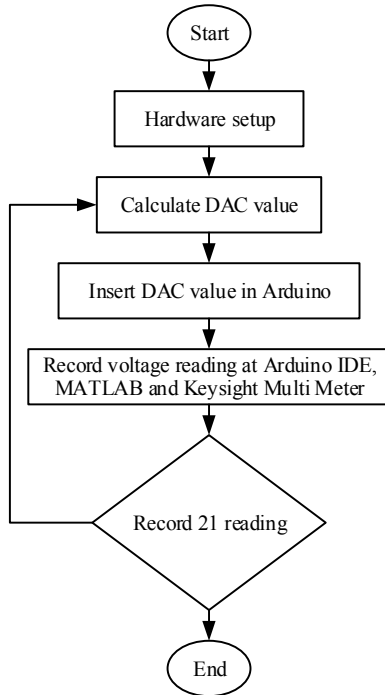


Fig. 2. Flowchart of the procedure to conduct the testing experiment.

First, the necessary hardware is prepared and connected as shown in Fig. 3.

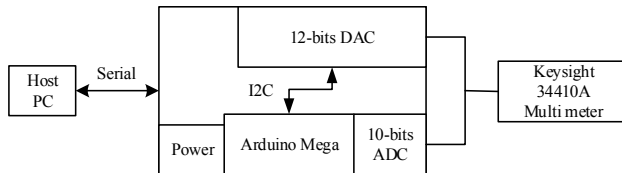


Fig. 3. Hardware setup of the testing experiment.

Next, the value sends by Arduino to the DAC can be calculated by finding the product of the desired percentage of instrument signal with DAC maximum out. The 12-bits MCP4725 DAC has a maximum output of 4096. The calculated value is then used to change the DAC out in the Arduino coding. The Arduino code is as follows:

```

#include <Wire.h>
#include <Adafruit_MCP4725.h>
#define voltsIn A0

Adafruit_MCP4725 dac; // constructor

void setup(void) {
  Serial.begin(9600);
  dac.begin(0x62); // The I2C Address
  Serial.println('Start');
}
  
```

```

void loop(void) {

  uint32_t dac_value = 4095; // output voltage of
  DAC
  int adcValueRead = 0;
  float voltageRead =

  dac.setVoltage(dac_value, false);
  adcValueRead = analogRead(voltsIn);
  voltageRead = (adcValueRead * 5.0) / 1024.0;

  Serial.println(voltageRead,3);
}
  
```

The output voltage will be displayed in Arduino serial monitor and send to MATLAB via serial communication. Keysight 34401A multimeter [19] is used to measure the output and input voltage of the DAQ. It is used to check the accuracy of the analog signal in the DAQ system because it is calibrated instrument from Keysight Technologies. The key sights 34401A has 6½ digit resolution and can perform ten type of measurement which is: DC/AC voltage, DC/AC current, 2- and 4-wire resistance, diode, continuity, frequency and period. This calibrated device has an accuracy of 0.0035% for DC measurement and 0.06% for AC measurement. The multimeter has 1000V maximum voltage input and 3A maximum current input. The measured value by this multimeter is used as ground truth for error analysis.

III. RESULTS AND ANALYSIS

The proposed DAQ system can be used for monitoring and control of process plant. For monitoring the sensor reading is feed to the host PC to display the sensor reading in real time and for control, the DAC will output analog signal to control the valve. As mentioned earlier, an experiment was conducted to verify the functionality of the proposed DAQ. The result for the testing of the DAQ system is tabulated in Table IV.

TABLE IV. TESTING EXPERIMENT RESULT

Instrument signal (%)	DAC	Output & Input		Keysight (Ground Truth) (V)
		MATLAB Reading (V)	Arduino Reading (V)	
0	0	0	0	1.182m
5	205	0.234	0.234	0.248
10	410	0.479	0.483	0.499
15	614	0.737	0.737	0.755
20	819	0.986	0.986	1.007
25	1024	1.24	1.24	1.261
30	1229	1.489	1.489	1.513
35	1433	1.738	1.743	1.765
40	1638	1.992	1.997	2.022
45	1843	2.241	2.241	2.271
50	2048	2.495	2.495	2.527
55	2252	2.739	2.739	2.777
60	2457	2.993	2.993	3.03
65	2662	3.242	3.242	3.286
70	2867	3.496	3.496	3.539
75	3071	3.750	3.750	3.793
80	3276	3.999	3.994	4.043
85	3481	4.243	4.243	4.295
90	3686	4.497	4.497	4.550
95	3890	4.751	4.746	4.802
100	4095	4.995	4.995	5.055

The values are then plotted in a graph to show the relation of each measurement.

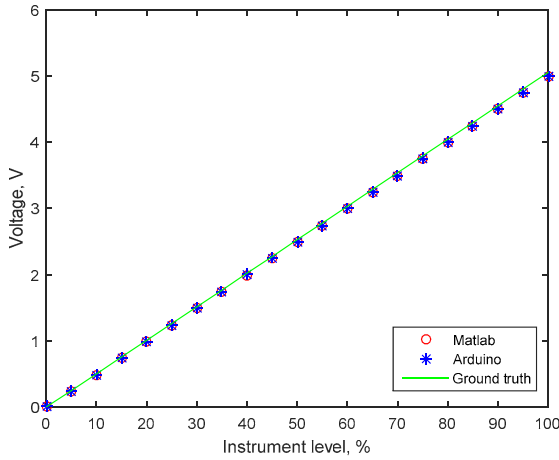


Fig. 4. The plot of output and input voltage against instrument level

Based on the graph, the DAQ system is free from non-linearity error and repeatability error. However, it has a sensitivity error. To measure the degree of the sensitivity error, the percentage difference between measured reading is compared with the ground truth. The percentage difference error was calculated according to:

$$\frac{R - G}{G} \times 100\% \quad (1)$$

where R is the voltage obtained from Arduino and MATLAB, and G is the voltage measured using Keysight 34401A multimeter. The result was then tabulated in Table V.

TABLE V. PERCENTAGE ERROR

Instrument signal (%)	Percentage Error (%)	
	MATLAB	Arduino
0	0	0
5	5.645	5.645
10	4.008	3.206
15	2.384	2.384
20	2.085	2.085
25	1.665	1.665
30	1.586	1.586
35	1.530	1.246
40	1.484	1.236
45	1.321	1.321
50	1.266	1.266
55	1.368	1.368
60	1.221	1.221
65	1.339	1.339
70	1.215	1.215
75	1.134	1.134
80	1.088	1.212
85	1.211	1.211
90	1.165	1.165
95	1.062	1.166
100	1.187	1.187

The accuracy of the DAQ system can be determined by analyzing the error. The system error can be obtained by averaging the values of percentage difference. The calculated errors are 1.693% and 1.748% for control and monitoring respectively. The proposed DAQ system has a very small sensitivity error. This proves the proposed DAQ system has a very high accuracy.

IV. CONCLUSION

In this paper, a low-cost DAQ for process automation and control is proposed. The DAQ is proven to be viable for used in process industry as based on the experimental results obtained, the proposed system can accurately generate an analog signal and read the analog input signal for process control and monitoring respectively. The proposed DAQ system has a high accuracy as the system error obtained over 21 measurement points are 1.693% and 1.748% respectively. The proposed DAQ can be implemented on a real process plant to verify the functionality of the DAQ and integrate with WirelessHART adapter technology [20,21,22] to enable remote monitoring and control for future work.

ACKNOWLEDGMENT

The authors acknowledge the support of Universiti Teknologi PETRONAS for carrying out this research through the funding of YUTP 0153AA-A74 grant and final year project budget scheme.

REFERENCES

- [1] I. Consulting, "Sensors Markets 2016, Development of Sensors Markets for the Non-Military, Open World Market until 2016," 2016.
- [2] A. Almalawi, A. Fahad, Z. Tari, A. Alamri, R. AlGhamdi and A. Y. Zomaya, "An Efficient Data-Driven Clustering Technique to Detect Attacks in SCADA Systems," in *IEEE Transactions on Information Forensics and Security*, vol. 11, no. 5, pp. 893-906, May 2016.
- [3] R. a Siddiqui, R. I. Grosvenor, and P. W. Prickett, "dsPIC-based Advanced Data Acquisition System for Monitoring , Control and Security Applications," *Int. Conf. Appl. Sci. Technol.*, pp. 293–298, 2015.
- [4] F. Li, X. Ji, X. Li, and K. Zhu, "DAQ architecture design of daya bay reactor neutrino experiment," *IEEE Trans. Nucl. Sci.*, vol. 58, no. 4 PART 1, pp. 1723–1727, 2011.
- [5] J. R. Blanco, F. J. Ferrero, M. Valledor, and J. Campo, "A Low-Cost Open-Source Data Acquisition System," pp. 1–6, 2014.
- [6] M. Abdallah, O. Elkeelany, and A. T. Alouani, "A low-cost stand-alone multichannel data acquisition, monitoring, and archival system with on-chip signal preprocessing," *IEEE Trans. Instrum. Meas.*, vol. 60, no. 8, pp. 2813–2827, 2011.
- [7] K. K. M. Rahman and M. Nasor, "Multipurpose low cost bio-daq system for real time biomedical applications," *2015 Int. Conf. Inf. Commun. Technol. Res. ICTRC 2015*, vol. 072, pp. 286–289, 2015.
- [8] S. Sharma and G. Kumar, "Low Cost Data Acquisition System for Students," *Int. J. Electron. Commun. Comput. Eng.*, vol. 2, no. 2, pp. 1–4, 2011.
- [9] S. Tehami, M. A. Khan, and O. Mazhar, "Hardware and software designing of USB based plug n play data acquisition device with C# and LabView compatibility," *2015 IEEE 21st Int. Symp. Des. Technol. Electron. Packag. SIITME 2015*, pp. 143–146, 2015.
- [10] M. Abdallah and O. Elkeelany, "A Survey on Data Acquisition Systems DAQ," *2009 Int. Conf. Comput. Eng. Inf.*, pp. 240–243, 2009.
- [11] User Guide NI USB-6001/6002/6003 Low-Cost DAQ USB Device. National Instrument, 2014.

- [12] CD-7021/CB7021P & CB-7022, CB-7024 User's Manual. Middleboro: Measurement Computing Corporation, 2000.
- [13] R. Acosta, Open Source Hardware, System Design and Management Program Thesis (S.M.), Massachusetts Institute of Technology, 2009
- [14] C. Harnett, "Open source hardware for instrumentation and measurement," in IEEE Instrumentation & Measurement Magazine, vol. 14, no. 3, pp. 34-38, June 2011.
- [15] A. E. Beltrán, "Low-cost acquisition and development board. An open source hardware proposal," Symposium of Signals, Images and Artificial Vision - 2013: STSIVA - 2013, Bogota, 2013, pp. 1-4.
- [16] N. Khera, H. Gill, G. Dodwani, N. Celly and S. Singh, "Remote Condition Monitoring of Real-Time Light Intensity and Temperature Data," Advances in Computing and Communication Engineering (ICACCE), 2015 Second International Conference on, Dehradun, 2015, pp. 3-6.
- [17] N. K. Lakkoju, S. Gudla and B. S. Mantravadi, "AVR-USB data acquisition," Electronics Computer Technology (ICECT), 2011 3rd International Conference on, Kanyakumari, 2011, pp. 35-39.
- [18] M. Abdallah, O. Elkeelany and A. T. Alouani, "A Low-Cost Stand-Alone Multichannel Data Acquisition, Monitoring, and Archival System With On-Chip Signal Preprocessing," in IEEE Transactions on Instrumentation and Measurement, vol. 60, no. 8, pp. 2813-2827, Aug. 2011.
- [19] 34401A Digital Multimeter Data Sheet. USA: Keysight Technologies, 2014.
- [20] T. Supramaniam, R. B. Ibrahim, T. D. Chung, and S. M. Hassan, "Towards the Development of WirelessHART Adaptor for Process Control Applications," presented at the 6th International Conference on Intelligent & Advanced Systems (ICIAS2016), Kuala Lumpur, 2016.
- [21] T. D. Chung, R. Ibrahim, V. S. Asirvadam, N. Saad, and S. M. Hassan, "Effect of Network Induced Delays on WirelessHART Control System," presented at the 6th International Conference on Intelligent & Advanced Systems (ICIAS2016), Kuala Lumpur, 2016.
- [22] S. M. Hassan, R. B. Ibrahim, N. B. Saad, V. S. Asirvadam, and T. D. Chung, "Implementation of Real-time WirelessHART Network for Control Application," presented at the 6th International Conference on Intelligent & Advanced Systems (ICIAS2016), Kuala Lumpur, 2016.